Beginner Course on Microbial Genomics: Applications & Research (2026)

Module 2.1

Post Assembly Profiling
Quality Assessment & Gene Content Analysis

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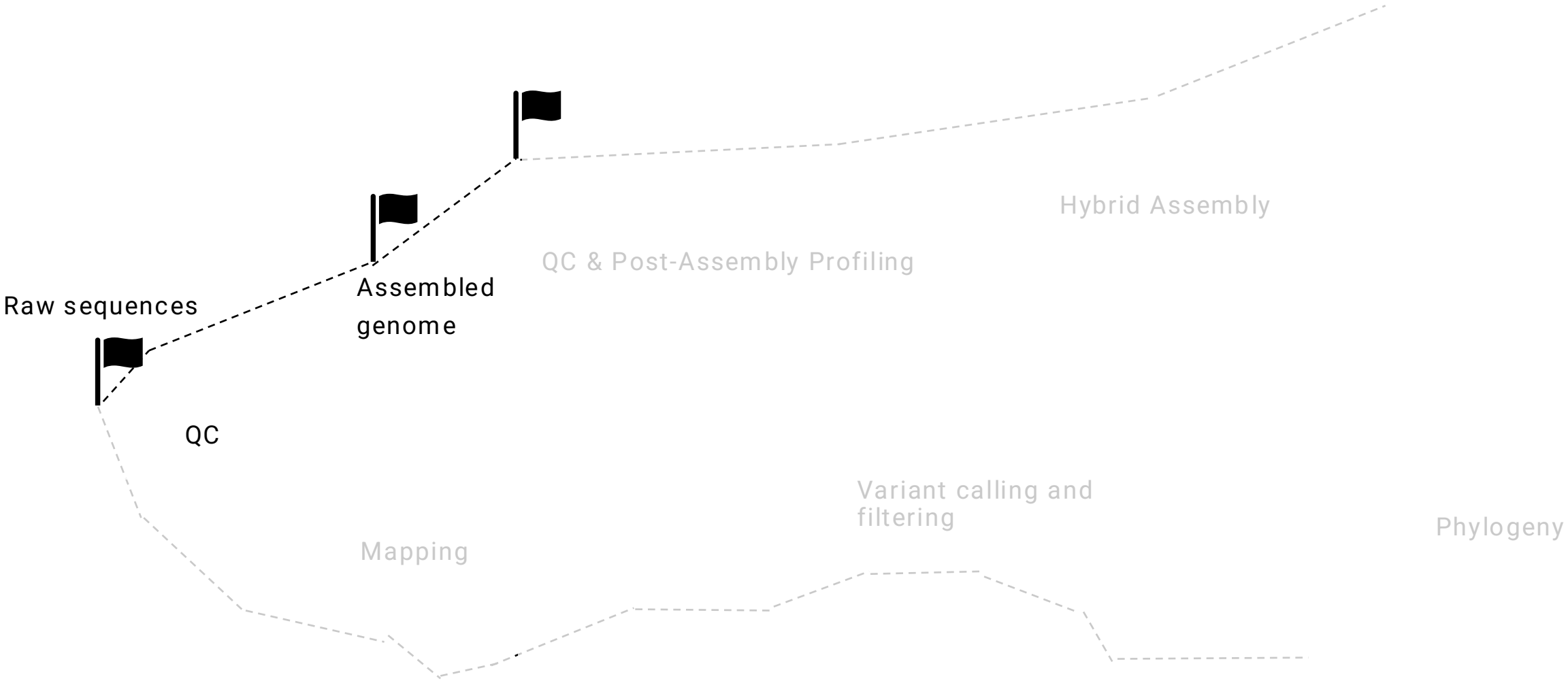
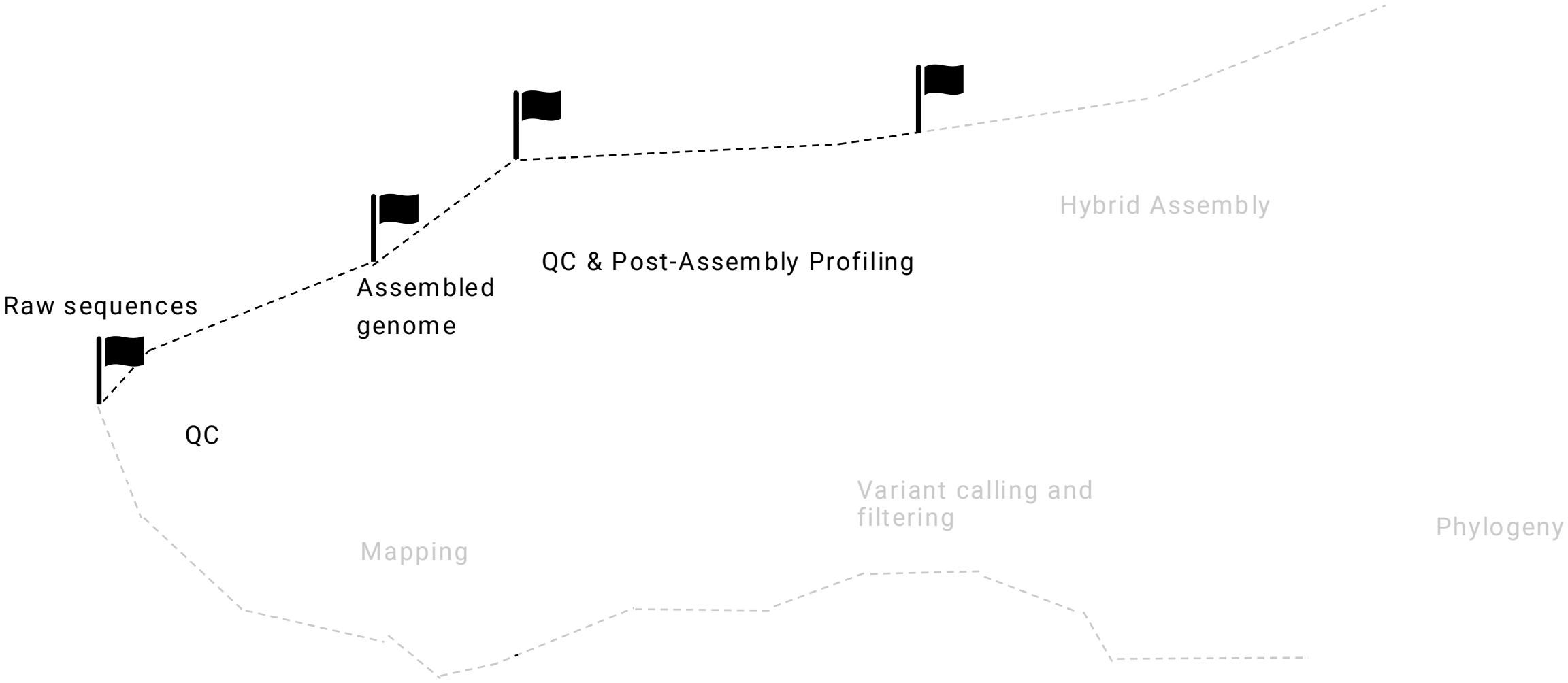
Our journeys



Our journeys



Agenda

I. **Theory**

1. Quality/Contamination Assessment
   a) Contiguity
   b) Completeness
   c) Contamination
2. Downstream Analysis Steps
   a) Multilocus Sequence Typing (MLST)
   b) AMR / Virulence genes identification

II. **Practical Command-line (CLI)**

1. checkm2
2. Kleborate

III. **Data Visualization and Manipulation with Bandage**

I. Theory

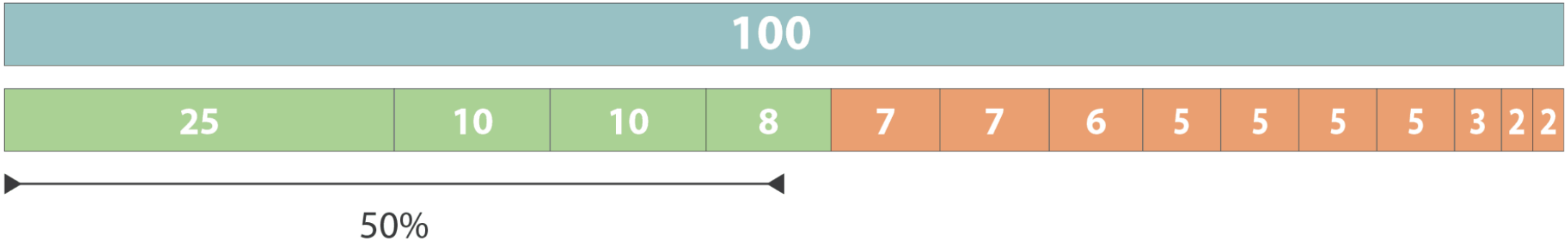
Theory

Quality/Contamination Assessment

- Contiguity : N50, Number of Contigs
- Completeness: Estimated Genome Size
- Contamination: SH index

Contiguity: N50 value

Definition: 50% of the total assembly size is contained in contigs of that length or longer

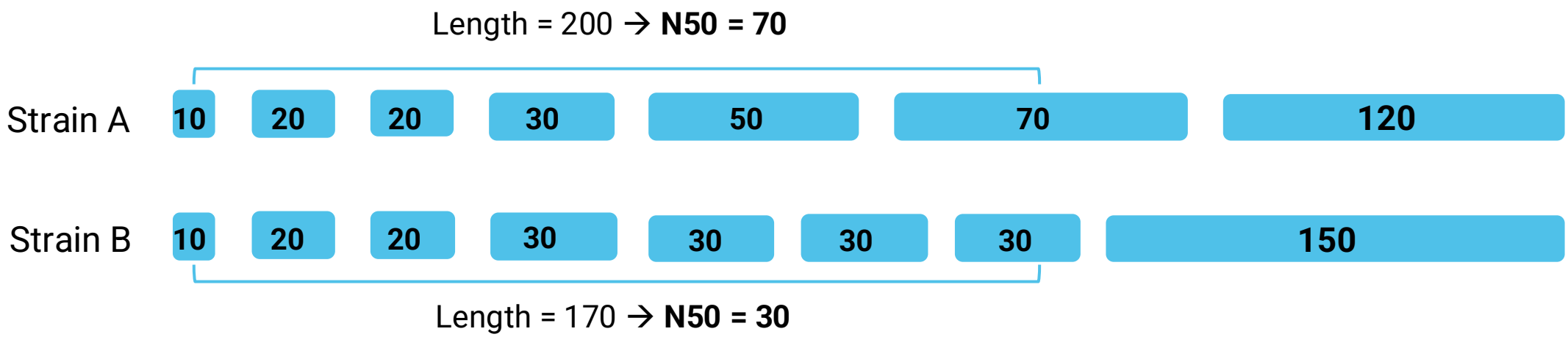


N50 value *(10.1371/journal.pcbi.1010752)*

N50 = 8

Contiguity: N50 value

Definition: 50% of the total assembly size is contained in contigs of that length or longer



Total length: 10 + 20 + 20 + 30 + 50 + 70 + 120 = **320**

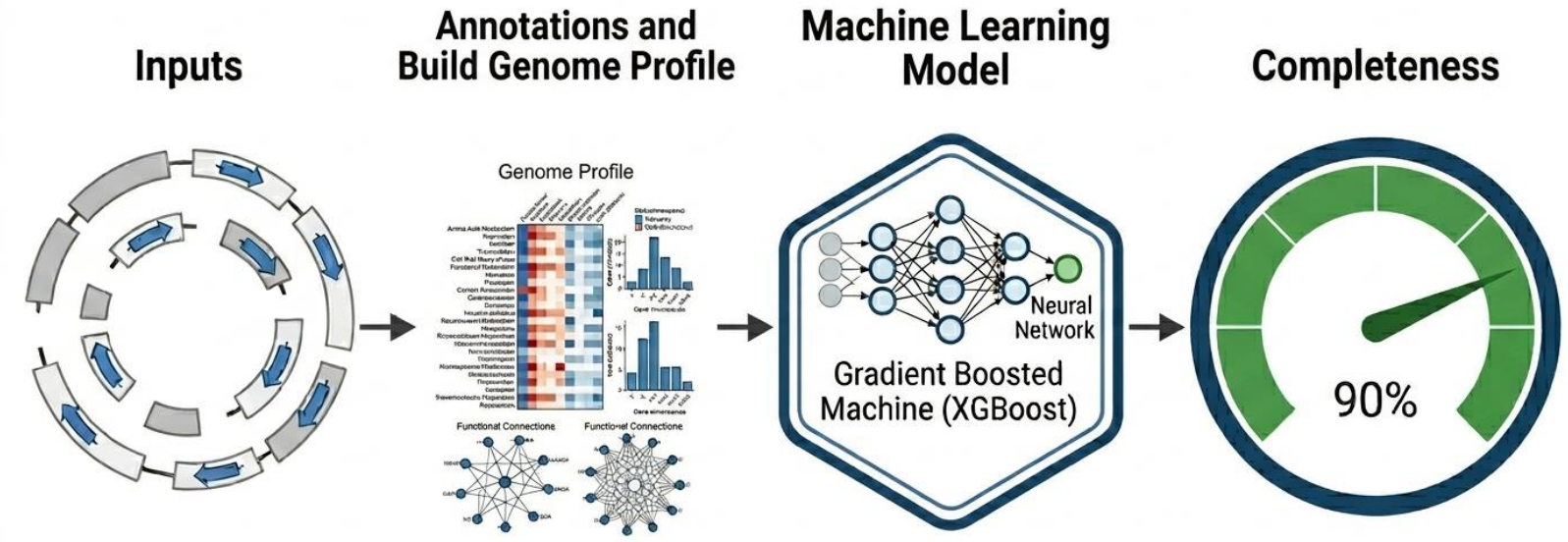
50% = 160

What is the N50 value of Strain A?

Which strain has better assembly quality?

Completeness: Estimated Genome Size (%)

Definition: percentage of the organism's total genome that is present in the assembly

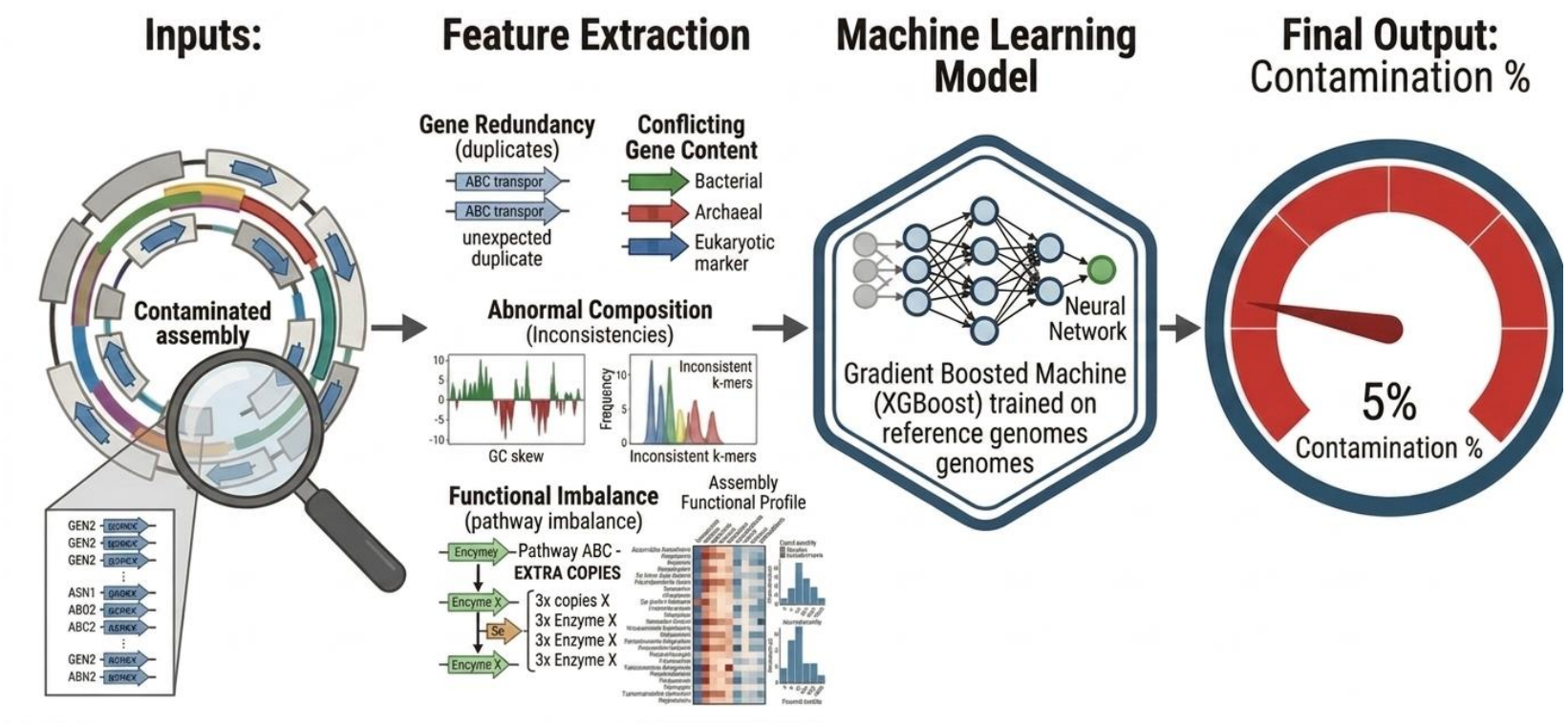


$$\frac{\text{Assembled genome size}}{\text{Estimated genome size}}$$

Recommend Quality Threshold - **Completeness: > 90%**

Contamination

Definition: percentage of sequences that originate from organisms other than the target genome



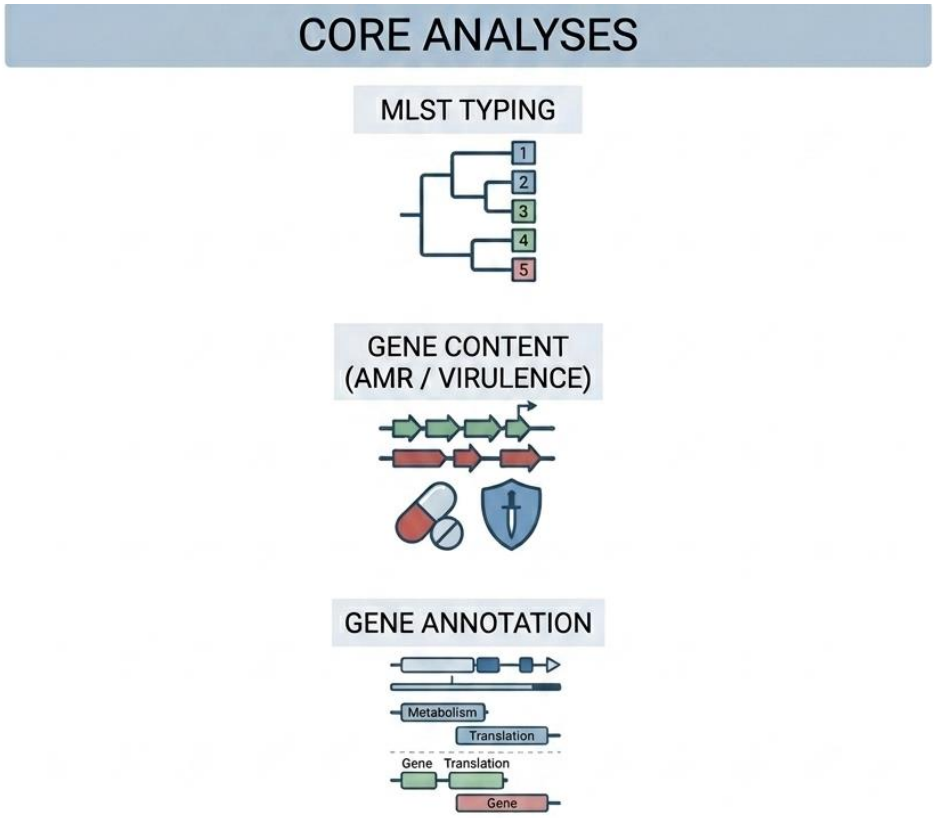
Recommend Quality Threshold - **Contamination: < 5%**

Theory

Downstream Analysis Steps

- MLST Typing
- Gene Content Identification

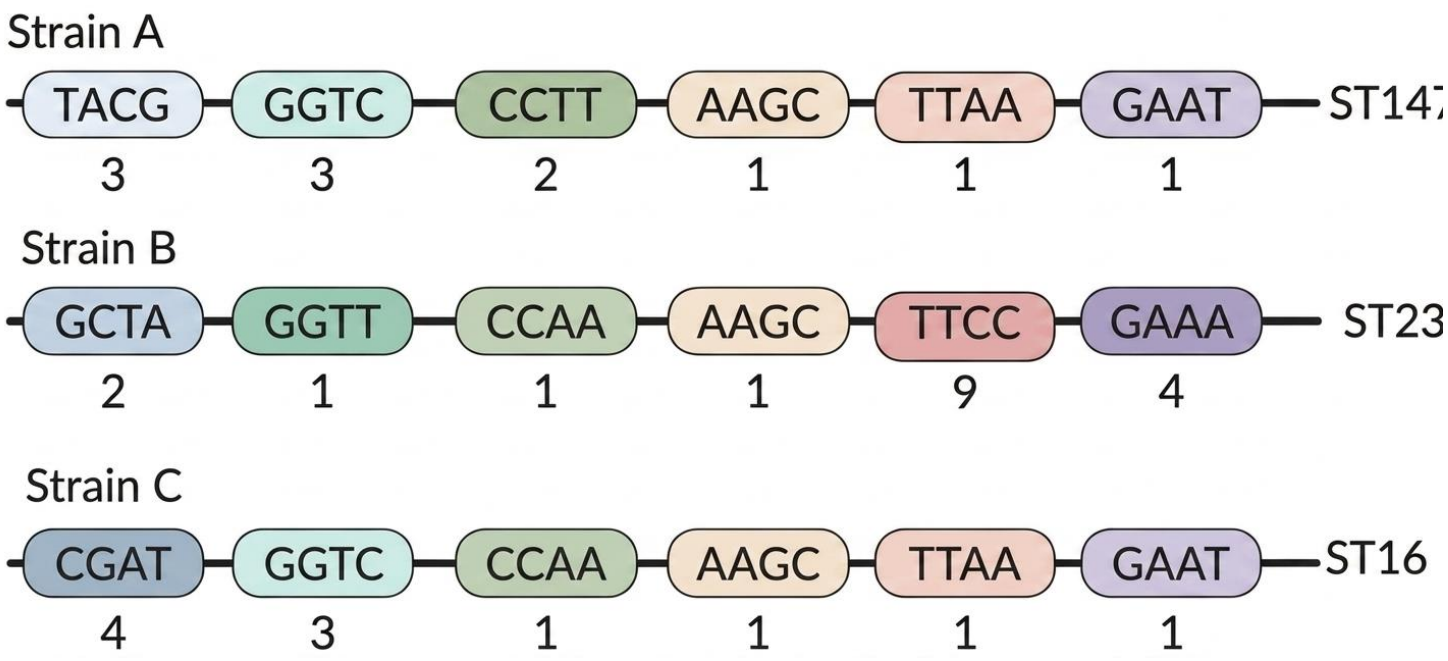
  (AMR/Virulence genes,…)
- Genome Annotation



Theory

MLST typing

Definition: **A housekeeping gene** is a gene that is always expressed on in bacterial cell, provides essential functions. **MLST** identifies a bacterial sequence type (ST) by analyzing the allelic profiles of several housekeeping genes.



PubMLST

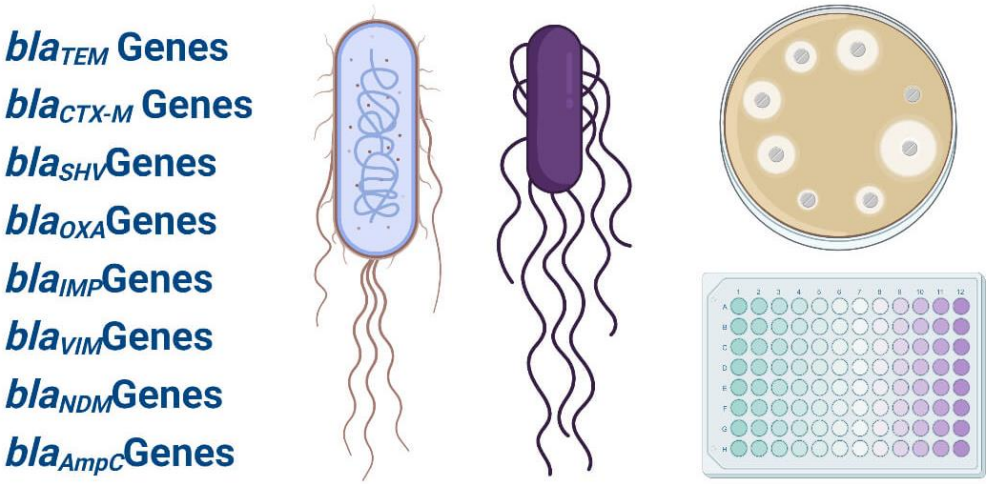
MLST Schemes

INSTITUT PASTEUR
BIGSdb-Pasteur

Theory

Gene Content Identification

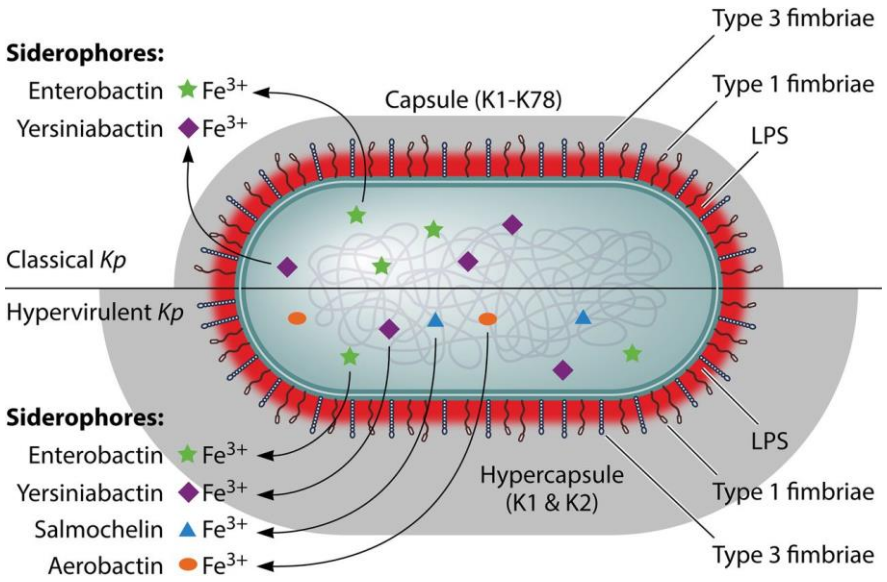
AMR



AMR genes in Enterobacteriaceae
(https://microbenotes.com/)

CARD

Virulence



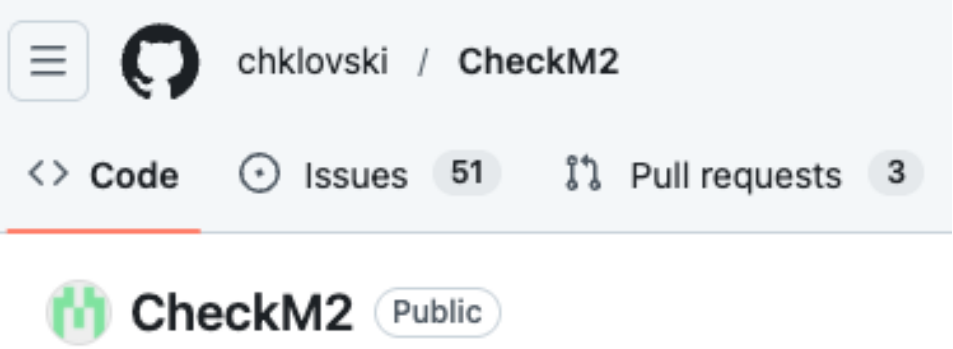
Virulence factors in classical and hypervirulent Kp
(10.1128/MMBR.00078-15)

VFDB *virulence factors of Pathogenic Bacteria*

II. Practical Command-line (CLI)

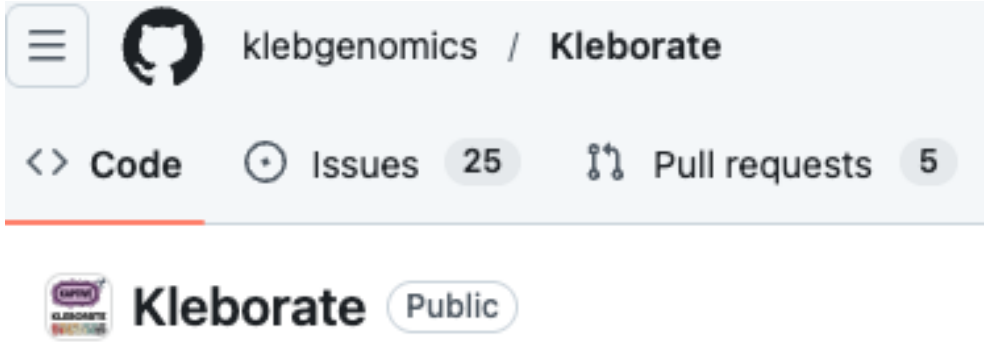
Quality/Contamination Assessment – checkm2

https://github.com/chklovski/CheckM2



Gene Content Indentification– kleborate

https://github.com/klebgenomics/Kleborate



III. Data Visualization and Manipulation

Bandage / BandageNG

THE END.

Thank you for your attention

Module 2.1: Post Assembly Profiling